The Ancient Art of War at Sea — Remake

Quick Reference Guide: building, running and command-line options

The remake is written in C with SDL3 and built with CMake. It runs on macOS, and also builds on Windows and Linux. No original game files are included: the remake loads its graphics, campaigns and data from **your own copy** of the original game at run time.

What you need

Tools	CMake 3.16 or later, a C11 compiler (Clang, GCC or MSVC) and SDL3. On macOS: `brew install cmake sdl3` (Homebrew).
Game files	From the original disk: `WAS.EXE`, `SM`, the SP files, `SO0`–`SO8`, `AD`, `B`, the S files (`S0`–`SA`), `SITLES` and `SDITLES`. Add `WAST.DAT` for the Tandy 1000's 16-colour graphics. `SCNF` is read if it is there. Upper or lower case doesn't matter.

Compiling

macOS

```
cd WAS_Remake
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build
```

This makes `build/AncientArtOfWarAtSea.app`. The app icon (the title page) is drawn during the build from your game files, found in `../WAS_Original_Files_3.5`. If they are elsewhere, add `-DWAS_GAME_DIR=/path/to/your/game` to the first `cmake` command. Without them the app still builds; it just has no icon in Finder. If Finder shows an old icon, run `touch build/AncientArtOfWarAtSea.app`.

Windows and Linux

The same two `cmake` commands, with SDL3 installed where CMake can find it (if it can't, add `-DCMAKE_PREFIX_PATH=/path/to/SDL3`). With Visual Studio, build with `cmake --build build --config Release`. The result is the `AncientArtOfWarAtSea` program in `build` (in `build/Release` for Visual Studio).

Tests (optional)

The test programs are in `WAS_Remake/test`. Build one with `./build.sh NAME` (with AddressSanitizer and UBSan; SDL3 from `pkg-config` or `SDL3_DIR`) and run it without a window by setting `SDL_VIDEODRIVER=dummy SDL_AUDIODRIVER=dummy`. Most take the game folder and an output folder. The test README lists each one and its arguments.

Running

```
open build/AncientArtOfWarAtSea.app                # macOS, from Finder or the terminal
build/AncientArtOfWarAtSea.app/Contents/MacOS/AncientArtOfWarAtSea ../WAS_Original_Files_3.5
```

On first launch the app asks for the folder that holds your game files and remembers it. It also looks in a `game` folder next to the app and in the current folder, and you can give the folder on the command line. Press **O** in the asset viewer to choose a different folder. Command-line options need the program inside the app bundle (`…/Contents/MacOS/AncientArtOfWarAtSea`).

Command-line options

```
AncientArtOfWarAtSea [--video=MODE] [HERC] [--log | --log=verbose] [game-folder]
AncientArtOfWarAtSea --screenshots OUTDIR [game-folder]
```

Option	What it does
`game-folder`	The folder with your original game files. Any argument that doesn't start with `-` (other than HERC) is taken as the folder.
`--video=MODE` `--video MODE`	The video mode to start in (CGA if not given). MODE is one of the names in the next table, in any case; anything else starts in CGA. Tab changes the mode while playing.
`HERC` `--herc`	Hercules monochrome, as the original's `WAS HERC`: the same as `--video=herc`.

Option	What it does
`--log` `--log=normal`	Write a debug log of each campaign played: the enemy commander's decisions and the reasons, your orders, the events and the sea battles.
`--log=verbose` `--log verbose`	The same, with every cell the enemy weighed, its route searches, its strength and port-value maps, every broadside and a table of all squadrons every 16 ticks.
`--screenshots OUTDIR`	Write every asset-viewer screen as BMP files for each video mode into OUTDIR, then quit (no window).

Video modes (--video=MODE)

MODE	Display	What you see
`cga`, `ega`, `vga`	CGA (default)	WAS.EXE's 320×200 in 4 colours: cyan, magenta and white; the scrolls in green, red and yellow. An EGA or VGA card showed the original exactly like this, so ega and vga start here
`pcjr`	IBM PCjr	The four colours WAS.EXE chose for each screen on a PCjr (the title page, the menus, the map and the battle each have their own)
`tandy`	Tandy 1000	The Tandy version's (WAST.EXE) 16-colour pictures and map, its yellow marker and coloured Info panel; the sea battle in light blue, brown, black and white, and boarding in 16 colours *
`herc`, `hercules`	Hercules	The CGA picture in monochrome, each pixel as two dots, in the middle of a 720×348 screen

* Needs `WAST.DAT`. Without it, Tab skips Tandy and `--video=tandy` starts in CGA. Tab goes through the modes in the order of this table. The game plays the same in every mode: only the pictures change.

> **Options Shoppe and the video modes:** in the PCjr mode the Options Shoppe's display line chooses which of each screen's colour sets the original used: *rgb*, *tv/composite* or *black/white* (shown in greys, as on a black-and-white monitor). In the other modes *black/white* turns the picture grey, and *rgb* and *tv/composite* look the same.

Examples

```
APP=build/AncientArtOfWarAtSea.app/Contents/MacOS/AncientArtOfWarAtSea   # macOS
$APP --video=tandy                   # the Tandy 1000's 16 colours
$APP --video=pcjr                    # as on an IBM PCjr
$APP HERC                            # as WAS HERC
$APP --log=verbose ~/Games/WAS       # a detailed log; game files in ~/Games/WAS
$APP --screenshots ~/Desktop/shots ../WAS_Original_Files_3.5
```

Keys that work everywhere

Key	Action
Tab	Next video mode (CGA, PCjr, Tandy, Hercules)
F2	The original monitor's 4:3 shape, or square pixels
⌘Enter on a Mac; Alt+Enter on Windows and Linux	Full screen
F12 (title page)	The asset viewer: every picture, the map tiles, the font, the map sprites and each campaign's map (Enter on a map plays it; Esc goes back)
Esc	Back up in the menus; during a game, quit to the title page (Y or N)
⌥S on a Mac (Option+S); Alt+S on Windows and Linux, during a game	Save the game on the data disk
⌘Q on a Mac	Quit

All the original game keys work as they did in the original (the map, the sea battle, boarding, the menus and the campaign editors); the remake's README lists them in full. In the original Tab scrolled the map up: - does the same.

Where the remake keeps its files

macOS: `~/Library/Application Support/AncientArtOfWarAtSea/Remake/` (on Windows and Linux, SDL's per-user data folder for the app).

File or folder	Contents
`gamedir.txt`	The folder with your original game files, as you chose it
`SCNF`	The Options Shoppe's settings, in the original's 29-byte layout
`DRIVE`	Which data disk is "in the drive"
`Data Disks/<name>/`	One folder per data disk (up to 12; the first, Data Disk 1, is made for you): `SITLES`, the `S` files of its campaigns, and its saved game `SS`. Campaigns and saved games are in the original's formats: to use a floppy's, copy its files into a new folder here.
`Logs/`	The `--log` files, one per campaign, named by date, time and campaign